

Data Journalism

Session 7: Writing with data

Simon Munzert

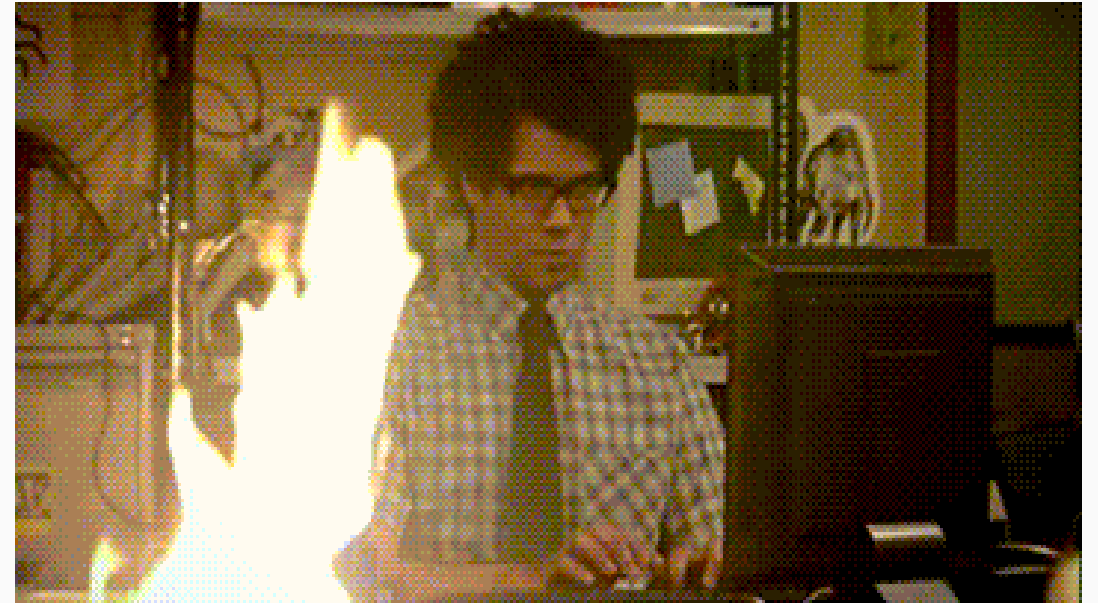
Hertie School | GRAD-E1493

Good writing

You are not being trained as journalists.

But if you analyze data, you will have to write, speak, or
present about data.

Today is about the craft of doing that well.



Today

1. What makes a data story?
2. Five lessons for writing with data
3. Ten bits of 1-minute advice

What makes a data story?

THE INFORMATION SOCIETY

<https://doi.org/10.1080/01972243.2021.1951415>



PERSPECTIVE

 OPEN ACCESS



Data storytelling is not storytelling with data: A framework for storytelling in science communication and data journalism

Sorin Adam Matei and Lucas Hunter

Brian Lamb School of Communication, College of Liberal Arts, Purdue University, West Lafayette, Indiana, USA

ABSTRACT

Storytelling teaches. All good storytelling is good teaching. Storytelling is a form of teaching and learning because it asks the readers or listeners to replace well-established explanations that are considered facts with new, unexpected ones. We always learn new things when listening to good stories. Good stories do so by violating expectations and surprising the listener or the reader. Surprise breeds suspense, which generates engagement, which is a catalyst for learning – even truer for scientific and data-driven stories. In this paper we show how understanding the nature of good stories by focusing on the novelty they introduce and assumption they violate helps us to do effective scientific work and tell excellent data-based stories. We start by providing definitions of “story,” “storytelling,” and “good stories.” We then outline a methodology for building stories and provide an illustrative example of an effective data-based story from the history of medicine.

ARTICLE HISTORY

Received 5 February 2021

Accepted 19 June 2021

KEYWORDS

Data story; data storytelling; expectation violation; learning; science communication; storytelling; teaching

Discussion

In pairs, discuss the following quotes from Matei and Hunter (2021):

When a story is good

A story is good because it satisfies a necessary and sufficient conditions; a good story always connects a cause to an effect (Schank 1991). Something leads to something else. Further, it is necessary the cause be unexpected. A good story neither states the obvious nor uses common sense as a universal yardstick (Dahlstrom 2014).

The role of surprise

The best stories employ surprise because surprise begets questions, which beget interest, which begets personal engagement, which leads to learning. All stories that use surprise work well, regardless of context or if they use numbers or not.

On scientific storytelling

Scientific storytelling, in turn, has a natural advantage over conventional storytelling because it consistently provides new, unexpected explanations. (...) By the nature of their practice, scientists can also be excellent storytellers. In fact, they are taught to think like storytellers.

On the role of novel explanations

When relying on data to make their point, stories which use a new theoretical model (one that better fits the facts while violating previous theories) have a higher chance to capture public attention. Thus, it is the job of the professional communicator, especially data scientists or journalists, to seek explanations which violate assumptions and known theories in a justified manner.

Scientific vs. journalistic writing

Scientific and journalistic writing are **two different genres**. They differ in their goals and audience, which has implications for the structure and style of the writing.

Compare

- **Language:** Scientific jargon vs. plain language
- **Structure:** Archetypical IMRD (Introduction, Methods, Results, Discussion) vs. the inverted pyramid (lede first, details later)
- **Title:** Describe the topic vs. Deliver the finding (or create suspense)

A mix of rather unexciting research article headlines

Little change in a changing landscape: Tracking exposure to untrustworthy news in Germany from 2017 to 2024. L Oswald, S Munzert Journal of Quantitative Description: Digital Media 6		2026
The zweitstimme forecast for the german federal election 2025: Coalition majorities and vacant districts C Erfort, LF Stoetzer, T Gschwend, E Koch, S Munzert, H Rajski PS: Political Science & Politics 59 (1), 48-54	4	2026
The 2025 report of the Lancet Countdown on health and climate change: climate change action offers a lifeline M Romanello, M Walawender, SC Hsu, A Moskeland, Y Palmeiro-Silva, ... The Lancet 406 (10521), 2804-2857	94	2025
An election forecasting model for subnational elections LF Stoetzer, C Erfort, H Rajski, T Gschwend, S Munzert, E Koch Electoral Studies 95, 102939	4	2025
Lost in moderation: How commercial content moderation apis over-and under-moderate group-targeted hate speech and linguistic variations D Hartmann, A Queslati, D Staufer, L Pohlmann, S Munzert, H Heuer Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems ...	26	2025
Citizen preferences for online hate speech regulation S Munzert, R Traunmüller, P Barberá, A Guess, JH Yang PNAS nexus 4 (2), pga032	11	2025
The Zweitstimme model: a forecast of the 2025 German federal election T Gschwend, S Munzert, H Rajski, LF Stoetzer		2025
The 2024 report of the Lancet Countdown on health and climate change: facing record-breaking threats from delayed action M Romanello, M Walawender, SC Hsu, A Moskeland, Y Palmeiro-Silva, ... The Lancet 404 (10465), 1847-1896	777	2024
Who's cheating on your survey? A detection approach with digital trace data S Munzert, S Ramirez-Ruiz, P Barberá, AM Guess, JH Yang Political Science Research and Methods 12 (2), 390-398	18 *	2024
The 2023 report of the Lancet Countdown on health and climate change: the imperative for a health-centred response in a world facing irreversible harms M Romanello, C Di Napoli, C Green, H Kennard, P Lampard, D Scamman, ... The Lancet 402 (10419), 2346-2394	1313	2023

Science or journalism?

Score: 0 / 0

Do Online Voter Guides Empower Citizens? Evidence from a Field Experiment with Digital Trace Data

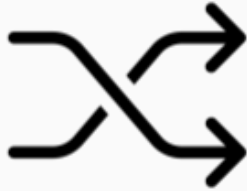


Journalism



Science

Common angles for data stories



Source [Paul Bradshaw, Online Journalism Blog](#)

Cost of rural crime hits £50 million in the UK



Rachel Martin

Aug 5, 2019, 5:49pm



Three quarters of driving test centres at maximum wait time



Almost all children in 73 areas of England live in low-income households

New official measures show levelling up attempts have failed to shift high levels of deprivation



Revealed: Israeli military's own data indicates civilian death rate of 83% in Gaza war

Scale: How big is an issue

Cost of rural crime hits £50 million in the UK



Rachel Martin

Aug 5, 2019, 5:49pm

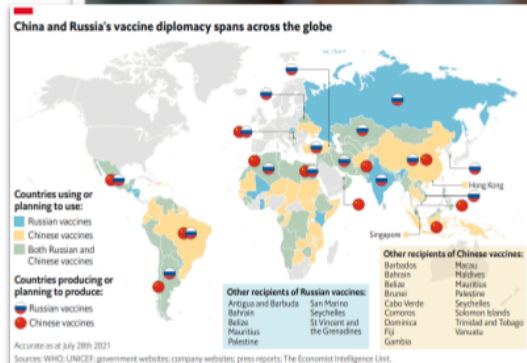


Three quarters of driving test centres at maximum wait time



Almost all children in 73 areas of England live in low-income households

New official measures show levelling up attempts have failed to shift high levels of deprivation



Revealed: Israeli military's own data indicates civilian death rate of 83% in Gaza war

School suspensions treble in 10 years in England

Popular music is getting sadder and angstier

Analyses of top-charting songs find a rise in blue moods

Rising racism blamed for collapse in number of foreign nurses coming to UK

Number of overseas nurses and midwives registered
between April and September was half that of a year ago



Coronavirus: Business rescue package has 'delayed the inevitable'

By Anna Khoo and Paul Bradshaw
BBC News

29 July



ANALYSIS

Revealed: Why yellow cards have been on the rise in the Premier League this season

The number of yellow cards for dissent have trebled in the Premier League across the opening 16 game weeks compared to the 2022/23 season; there have been 80 cautions for such an offence compared to 24 last term

Change: Things getting better/up or worse/down

School suspensions treble in 10 years in England

Popular music is getting sadder and angstier

Analyses of top-charting songs find a rise in blue moods

Rising racism blamed for collapse in number of foreign nurses coming to UK

Number of overseas nurses and midwives registered between April and September was half that of a year ago



Coronavirus: Business rescue package has 'delayed the inevitable'

By Anna Khoo and Paul Bradshaw
BBC News

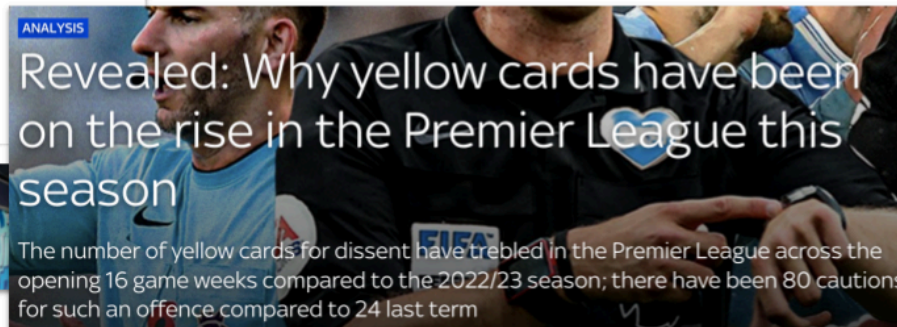
29 July



ANALYSIS

Revealed: Why yellow cards have been on the rise in the Premier League this season

The number of yellow cards for dissent have trebled in the Premier League across the opening 16 game weeks compared to the 2022/23 season; there have been 80 cautions for such an offence compared to 24 last term



Sad songs say so much

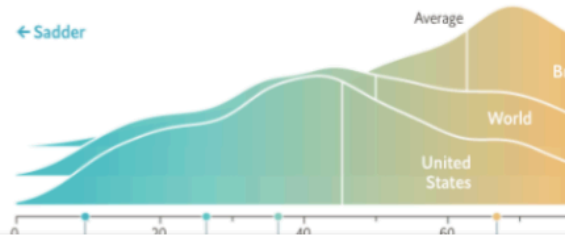
Data from Spotify suggest that listeners are gloomiest in February

Around the world, the most popular tunes this month will

→ Some countries listen to happier music than others

Distribution of tracks streamed*, by mood

← Sadder



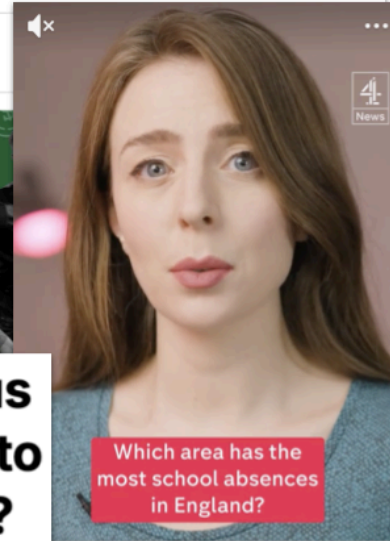
m+ EXCLUSIVE Britain's most dangerous train stations revealed as crimes soar to all-time high... so is YOURS on the list?

By **OLIVER PRICE**, ASSISTANT DATA EDITOR

FIFA: Saudi Arabia third biggest spenders on international transfers in 2023



WARREN MENEZES
30 JAN 2024



Ranking: The best/worst/outliers

Sad songs say so much

Data from Spotify suggest that listeners are gloomiest in February

Around the world, the most popular tunes this month will

→ Some countries listen to happier music than others

Distribution of tracks streamed*, by mood

← Sadder

United States

World

Average

FIFA: Saudi Arabia third biggest spenders on international transfers in 2023

WARREN MENEZES

30 JAN 2024

Which area has the most school absences in England?

m+ EXCLUSIVE Britain's most dangerous train stations revealed as crimes soar to all-time high... so is YOURS on the list?

By OLIVER PRICE, ASSISTANT DATA EDITOR

Shot by cops, thwarted by judges and geography

U.S. courts show wide regional disparities in granting qualified immunity, the controversial doctrine now under fire for protecting officers accused of excessive force.

Positive assessment rates for autism vary widely between parts of England

Charity says findings on evaluations of children and adults suggest different approaches and are 'deeply worrying'



Social Care in Yorkshire

Doncaster Council
70% rejected

Calderdale Council
18% rejected

Scottish data shows big variations in school exclusions

World Menopause Day: HRT shortages in parts of England leaves women to 'suffer in silence'

Women are facing a 'postcode lottery' in accessing Hormonal Replacement Therapy (HRT) medication and menopause treatment in England, with some waiting as long as 18 months for NHS treatment, campaigners warn.

Variation: Things are different across space/subgroups

Shot by cops, thwarted by judges and geography

U.S. courts show wide regional disparities in granting qualified immunity, the controversial doctrine now under fire for protecting officers accused of excessive force.

Positive assessment rates for autism vary widely between parts of England

Charity says findings on evaluations of children and adults suggest different approaches and are 'deeply worrying'

Social Care in Yorkshire

Doncaster Council
70% rejected

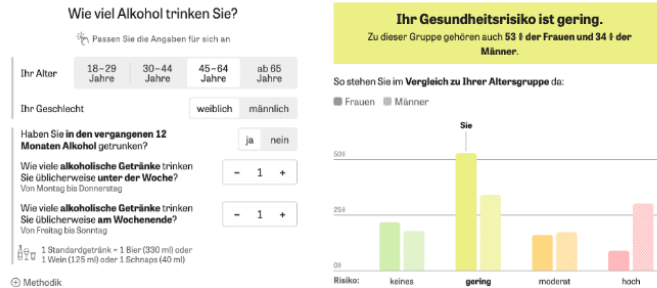
Calderdale Council
18% rejected

Scottish data shows big variations in school exclusions

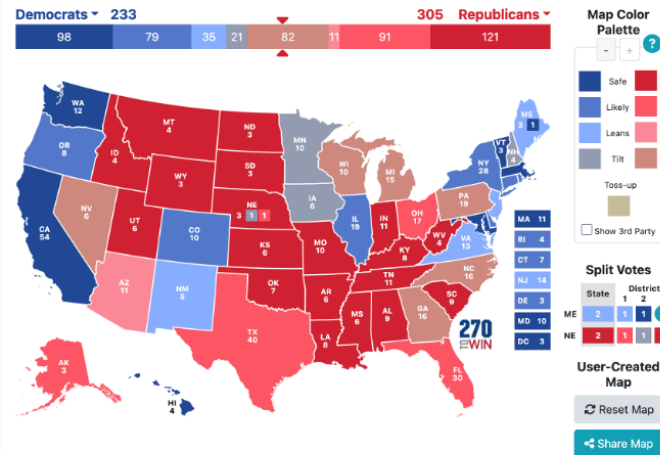
World Menopause Day: HRT shortages in parts of England leaves women to 'suffer in silence'

Women are facing a 'postcode lottery' in accessing Hormonal Replacement Therapy (HRT) medication and menopause treatment in England, with some waiting as long as 18 months for NHS treatment, campaigners warn.

Wie riskant ist Ihr Alkoholkonsum?



It will take 270 electoral votes to win the 2028 presidential election. Click states on this interactive map to create your own 2028 election forecast. **Create a specific match-up** by clicking the party and/or names near the electoral vote counter. Use the buttons below the map to share your forecast or embed it into a web page.



How Much Hotter Is Your Hometown Than When You Were Born?

As the world warms because of human-induced climate change, most of us can expect to see temperatures hit 32 degrees Celsius or higher. See how your hometown compares to how much hotter it may get.

Durham, North Carolina

Weißer Weihnachten in ...?

Alle Jahre wieder fragen wir uns, ob es an Heiligabend und an den Feiertagen schneien wird. Geben Sie Ihren Wohnort ein und sehen Sie, wann es bei Ihnen in den letzten 70 Jahren weiße Weihnachten gab. Das dekorative Ergebnis können Sie herunterladen, ausdrucken und vielseitig einsetzen: als Poster, Last-Minute-Geschenkpapier oder Grußkarte.

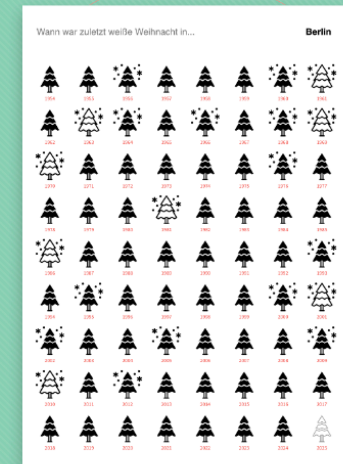
Geben Sie eine Stadt ein...

Berlin, Köln, Hamburg, Erfurt, Braunschweig, Essen, Winterberg

Download

Drucken Sie die Grafik aus, beispielsweise als:

Poster, Geschenkpapier, Postkarte

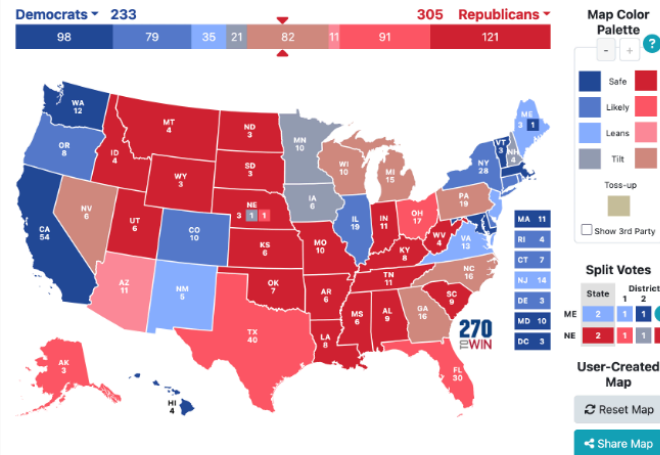


gar kein Schnee, wenigstens 1 x Schnee, alle Tage Schnee

Exploration: Tools, simulators, interactivity, personalization



It will take 270 electoral votes to win the 2028 presidential election. Click states on this interactive map to create your own 2028 election forecast. Create a specific match-up by clicking the party and/or names near the electoral vote counter. Use the buttons below the map to share your forecast or embed it into a web page.



How Much Hotter Is Your Hometown Than When You Were Born?

As the world warms because of human-induced climate change, most of us can expect to see temperatures hit 32 degrees Celsius or higher. See how your hometown compares to how much hotter it may get.

Durham, North Carolina

Weißer Weihnachten in ...?

Alle Jahre wieder fragen wir uns, ob es an Heiligabend und an den Feiertagen schneien wird. Geben Sie Ihren Wohnort ein und sehen Sie, wann es bei Ihnen in den letzten 70 Jahren weiße Weihnachten gab. Das dekorative Ergebnis können Sie herunterladen, ausdrucken und vielseitig einsetzen: als Poster, Last-Minute-Geschenkpapier oder Grußkarte.

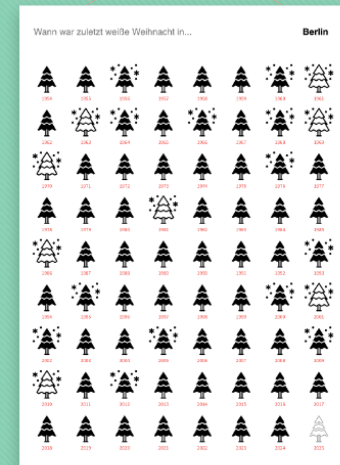
Geben Sie eine Stadt ein...

Berlin, Köln, Hamburg, Erfurt, Braunschweig, Essen, Winterberg

Download

Drucken Sie die Grafik aus, beispielsweise als:

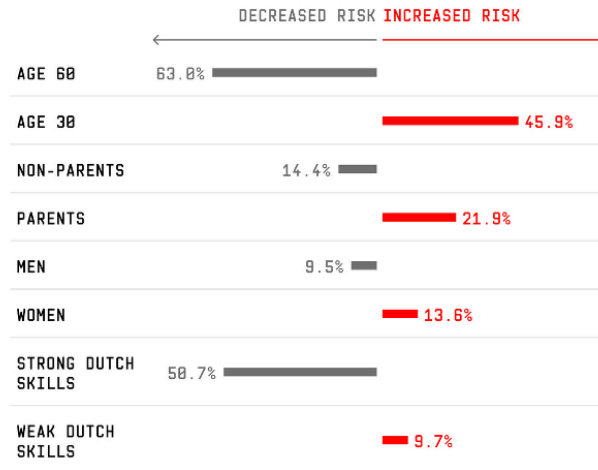
Poster, Geschenkpapier, Postkarte



Inside the Suspicion Machine

Obscure government algorithms are making life-changing decisions about millions of people around the world. Here, for the first time, we reveal how one of these systems works.

DIFFERENT TRAITS, DIFFERENT RISK SCORES

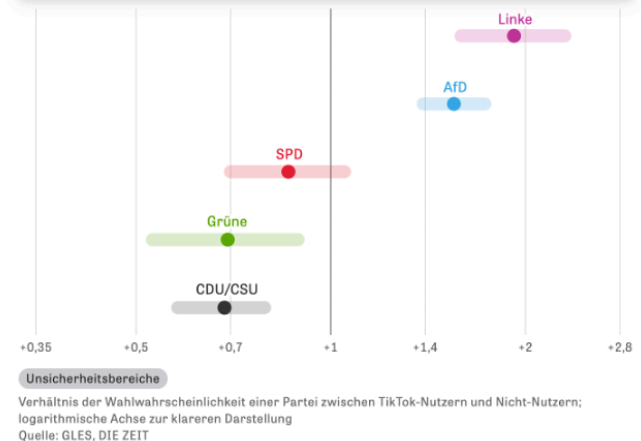


What's driving up burger prices?

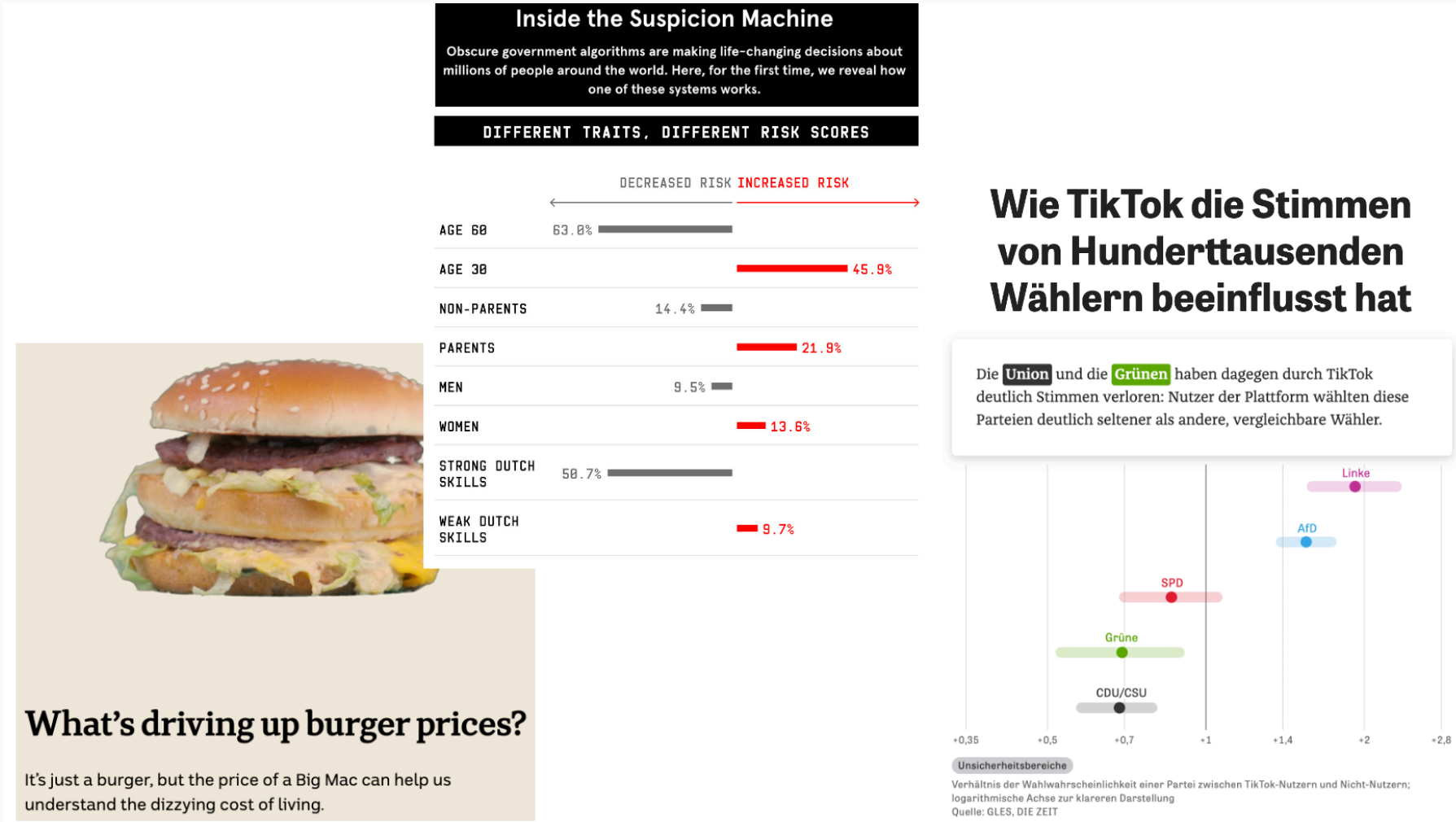
It's just a burger, but the price of a Big Mac can help us understand the dizzying cost of living.

Wie TikTok die Stimmen von Hunderttausenden Wählern beeinflusst hat

Die **Union** und die **Grünen** haben dagegen durch TikTok deutlich Stimmen verloren: Nutzer der Plattform wählten diese Parteien deutlich seltener als andere, vergleichbare Wähler.



Relationships: How things are (causally) connected (and why)



MAR. 31, 2020, AT 12:16 PM

Why It's So Freaking Hard To Make A Good COVID-19 Model

SEP. 30, 2019, AT 7:29 AM

How A Big Enough News Story — Like Impeachment — Could Warp The Polls

How the Pinellas Sheriff's Office boosts its rape stats without solving cases

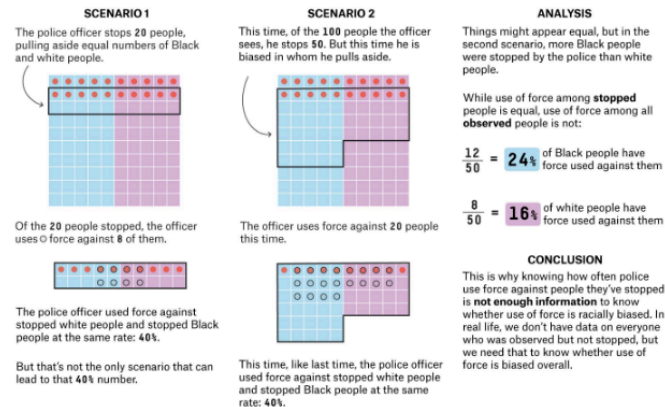
"None of these cases I reviewed were properly or thoroughly investigated," said a retired criminal investigator, who reviewed 11 of the cleared cases for the Times.



Why Statistics Don't Capture The Full Extent Of The Systemic Bias In Policing

How numbers that appear equitable can obscure bias

Let's say a police officer is patrolling the street, looking for people with contraband. The officer sees 100 people, some of whom have contraband on their person. Say the crowd is evenly split between Black and white people.



Data: Bad data, good data, hey we got the data stories

MAR. 31, 2020, AT 12:16 PM

Why It's So Freaking Hard To Make A Good COVID-19 Model

SEP. 30, 2019, AT 7:29 AM

How A Big Enough News Story — Like Impeachment — Could Warp The Polls

How the Pinellas Sheriff's Office boosts its rape stats without solving cases

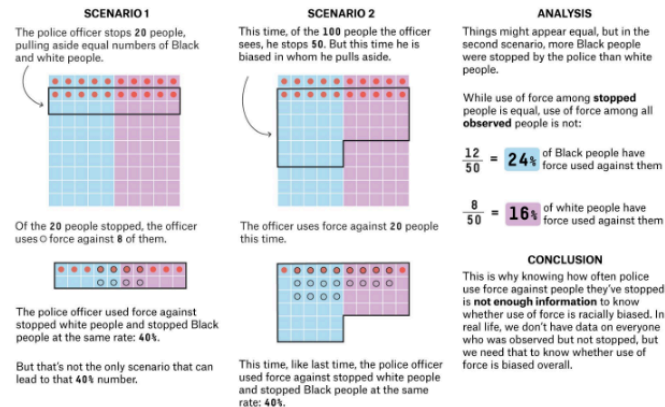
"None of these cases I reviewed were properly or thoroughly investigated," said a retired criminal investigator, who reviewed 11 of the cleared cases for the Times.



Why Statistics Don't Capture The Full Extent Of The Systemic Bias In Policing

How numbers that appear equitable can obscure bias

Let's say a police officer is patrolling the street, looking for people with contraband. The officer sees 100 people, some of whom have contraband on their person. Say the crowd is evenly split between Black and white people.



Five lessons for writing with data

Lesson 1: Start from the one-sentence claim

The lesson

- Before you write a single paragraph, write your claim in **one sentence**. If you can't, you don't yet know what you're writing about.
- The one-sentence claim becomes your **organizing principle** — the headline, the lede etc. all point at it.
- If your claim changes halfway through the draft, fine — but rewrite the opening to match.

The headline

- In the end, your claim can be the headline.
- In any case, **a headline is a claim, not a topic**. "Housing in Berlin" is not a headline. "Berlin rents rose faster last year than in any year since 1991" is.
- Good test: can your reader tell, from the headline alone, what the story is? If not, rewrite.

A simple template

"[In population / context], [what happened / is happening], [by how much / compared to what]."

Examples

✗ Topic headline	✓ Finding headline
"A look at EU antibiotic use"	"Europe is winning the antibiotic battle — unevenly"
"Gender in German startups"	"German startups led by women raise half as much"
"Housing in Berlin"	"Berlin's rents rose 130% in a decade"

The headline is probably the last thing that you write — but it should be the first thing you think about.

Exercise: Write the claim

You have this finding:

A new analysis of Germany's startup ecosystem, based on data from the Federal Statistical Office and Crunchbase covering roughly 500 venture-backed companies founded between 2019 and 2024, sheds light on how founder demographics may relate to early-stage fundraising outcomes. The study, which considers firms across sectors including fintech, climate tech, and enterprise software, examines a range of variables including founding team size, geographic location, and prior work experience. Among its findings, the analysis indicates that, all else being roughly equal, founders under the age of 30 tend to close their Series A rounds at valuations that translate to approximately €1.2 million less on average than their counterparts aged 30 and above — a gap that persists even after accounting for the fact that the two groups showed broadly comparable revenue growth trajectories in the twelve months leading up to the round. Industry observers have long debated whether age serves as a proxy for experience, network depth, or investor bias, and while this study does not resolve that question, the results arguably add a quantitative data point to an ongoing conversation.

Task (3 minutes, individually)

Write the **one-sentence claim** you would put at the top of the article. Aim for:

- A verb that does work (not "examines", not "looks at")
- A specific number
- A comparison or a "but"

Exercise: Rewrite these headlines

Topic → Finding (5 minutes)

Rewrite each topic headline as a **finding headline**. You will need to invent plausible numbers — that's fine, it's an exercise in form.

1. *"A study of commuting patterns in German cities"*
2. *"An analysis of climate spending in the EU budget"*
3. *"Women in German politics"*
4. *"AI adoption in small businesses"*

Lesson 2: Build a solid structure — the inverted pyramid

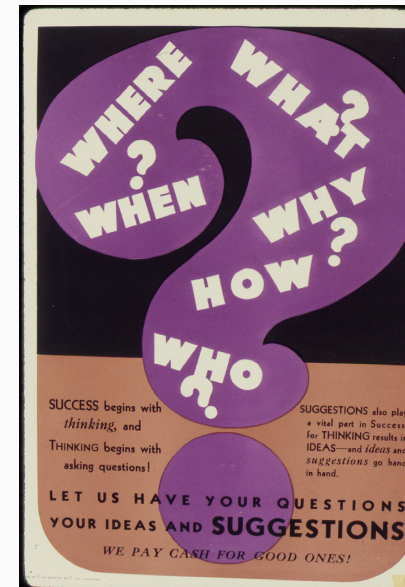
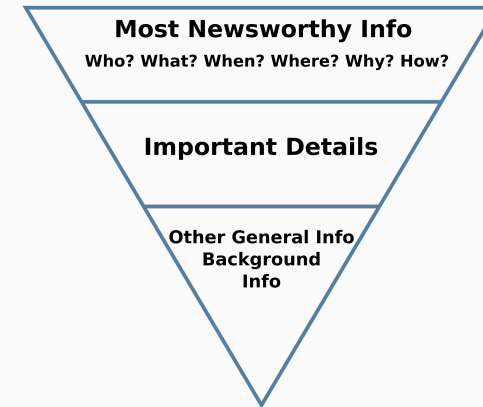
The lesson

- **Most important thing first.** Less important after. Caveats come last.
- Readers drop off paragraph by paragraph. Write so that whenever they stop, they still got the point.

The classic order

1. **Lede** — the finding addressing the five Ws
2. **Nut graf** — why it matters in a NUTshell
3. **Evidence** — the key numbers
4. **Context** — comparisons, history
5. **Voices** — quotes, alternative views
6. **Methods & caveats** — how we know

More freestyle narrative arcs work too — but only if you have a reason to use one.



Exercise: Put the paragraphs in order

Here are six paragraphs from a story — shuffled. Put them in inverted-pyramid order.

- (a) *"The study used anonymised tax records from 2015 to 2023, controlling for age, sector, and firm size. Its authors caution that cross-country comparisons remain difficult."*
- (b) *"Germany's gender pay gap narrowed for the first time in a decade last year — and the driver was not what policymakers expected."*
- (c) *"The shift was driven almost entirely by wage growth among women in the public sector, where new collective bargaining agreements raised base salaries by 11% on average."*
- (d) *"'This is not a story about firms finally paying women fairly,' said Prof. Anna Weber of the DIW. 'It's a story about which sector women happen to be in.'"*
- (e) *"The gap now stands at 15.7%, down from 18.3% in 2022 — the sharpest single-year drop since Germany began tracking the measure."*
- (f) *"If the finding holds up, it suggests that private-sector pay practices have scarcely budged — and that the headline progress masks deeper stagnation."*

Exercise: Put the paragraphs in order

Here are six paragraphs from a story — shuffled. Put them in inverted-pyramid order.

- (a) *"The study used anonymised tax records from 2015 to 2023, controlling for age, sector, and firm size. Its authors caution that cross-country comparisons remain difficult."*
- (b) *"Germany's gender pay gap narrowed for the first time in a decade last year — and the driver was not what policymakers expected."*
- (c) *"The shift was driven almost entirely by wage growth among women in the public sector, where new collective bargaining agreements raised base salaries by 11% on average."*
- (d) *"'This is not a story about firms finally paying women fairly,' said Prof. Anna Weber of the DIW. 'It's a story about which sector women happen to be in.'"*
- (e) *"The gap now stands at 15.7%, down from 18.3% in 2022 — the sharpest single-year drop since Germany began tracking the measure."*
- (f) *"If the finding holds up, it suggests that private-sector pay practices have scarcely budged — and that the headline progress masks deeper stagnation."*

Good order: b → f → e → c → d → a. *Do you agree?*

Lesson 3: Write a strong lede

The lesson

- The **first sentence** has one job: make the reader stay for the second.
- Two structures work: (a) **direct lede** — state the finding. (b) **scene-setting lede** — start with a person or place, then pivot by paragraph 2.
- Both pieces need a **nut graf** — the paragraph that says *"this matters because..."* — within the first three paragraphs.
- What does *not* work: the **background lede** ("For years, researchers have debated..."). Readers leave.

The litmus test

Cover your first paragraph. Can the reader still tell what the story claims? If yes, your lede buries the finding. This is (often) bad.

Good ledes: Examples

Accountability

"Doctors in New Jersey have received at least **\$2.1 billion** from drug companies over the past decade, according to federal records — payments that critics say skew prescribing decisions toward the companies' own products."

Puzzle in need of explanation

"The United States is dying faster than it used to. After decades of gains, American life expectancy fell in **2015** — and it has barely recovered since."

Personal

"If you were born in Berlin-Mitte in 1990, you have watched your rent rise **more than 130%**. If you were born in Gelsenkirchen, it has barely moved."

Exercise: Compare two ledes

The story

A study finds that German medical students from rural backgrounds are 40% less likely to become doctors in rural areas than their urban peers — the opposite of long-standing policy assumptions.

Lede A

"For decades, German medical policy has assumed that recruiting students from rural areas will improve doctor shortages in the countryside. A new analysis from the Robert Koch Institute, published this week in the journal Gesundheitswesen, examined records of 12,000 medical school graduates across 16 federal states between 2005 and 2020..."

Lede B

"Germany's rural doctor shortage has a surprising source: rural medical students. A new analysis of 12,000 graduates finds that students from rural backgrounds are 40% less likely to practise in the countryside than their urban classmates — overturning a central assumption of national health policy."

Discuss: Which lede do you find stronger? Why?

Lesson 4: Make every number concrete

The lesson

- **Raw numbers don't land — they need anchors.** "€2 billion" is abstract. "€2 billion — about double the yearly budget of the city of Potsdam" is concrete.
- Use comparisons: **per person, per day, as a share of, a multiplier of something familiar.**
- **One or two precise numbers per paragraph.** Round for prose, keep precision in tables.
- Avoid false precision: "4,273 died" sounds scientific but often isn't. "More than 4,000" is usually more honest.
- Never use a unit readers can't picture (hectares, petabytes, basis points) without translating.

Exercise: Make the numbers concrete

Rewrite each sentence to make the number land (5 minutes)

1. *"Germany spent €47.3 billion on defence in 2024."*
2. *"The model's classification accuracy was 83.2%."*
3. *"The study sampled 2,147 households."*
4. *"The company's carbon footprint was 1.4 million tonnes of CO₂."*

Share one with the group

What **anchor** did you pick? (per capita? share of GDP? equivalent in cars, flights, trees?) Why that one?

Lesson 5: Revise — cut hedging, jargon, filler

The lesson

- Your first draft is not your final draft.
- Cut: *"it should be noted that"*, *"arguably"*, *"somewhat"*, *"a number of"*, *"in order to"*, *"at this point in time"*.
- Replace jargon with plain words: *utilize* → *use*; *methodology* → *method*; *incentivize* → *pay*.
- Shorten every sentence by 20%. Then do it again.
- **Read the draft aloud.** If you stumble, the reader will too.
- Be honest about uncertainty — but honestly, not evasively. *"We don't know"* is better than *"it remains to be seen"*.

Simple rule

If you can delete a word without changing the meaning, delete it.

Exercise: Edit this paragraph

Cut it by at least 30%. Keep the meaning. (7 minutes)

"Drawing on administrative data from Destatis and the Federal Employment Agency covering roughly 45,000 small and medium-sized enterprises (SMEs) in rural districts between 2014 and 2022, this analysis finds evidence of what appears to be a notable association between the expansion of broadband internet access and hiring activity at the firm level. Specifically, a 10-percentage-point increase in household broadband coverage within a district is associated with an approximately 3.4 percent higher annual hiring rate among local SMEs, with the effect being somewhat more pronounced for firms with fewer than 50 employees. It should be noted that these findings are correlational in nature and do not necessarily establish a causal relationship, and a number of methodological caveats apply, particularly regarding unobserved regional heterogeneity."

Compare with your neighbour

Whose cut was tighter? What did you each delete first? Any words you **kept** that your neighbour cut — and why?

Exercise: Edit this paragraph

Reduce the following paragraph to 2 sentences. Keep the key messages (7 minutes)

"Drawing on administrative data from Destatis and the Federal Employment Agency covering roughly 45,000 small and medium-sized enterprises (SMEs) in rural districts between 2014 and 2022, this analysis finds evidence of what appears to be a notable association between the expansion of broadband internet access and hiring activity at the firm level. Specifically, a 10-percentage-point increase in household broadband coverage within a district is associated with an approximately 3.4 percent higher annual hiring rate among local SMEs, with the effect being somewhat more pronounced for firms with fewer than 50 employees. It should be noted that these findings are correlational in nature and do not necessarily establish a causal relationship, and a number of methodological caveats apply, particularly regarding unobserved regional heterogeneity."

Compare with your neighbour

Which version is more on point? What did you each delete first? Any words you **kept** that your neighbour cut — and why?

Sample answer: "New analysis of German federal data links rural broadband access to company hiring. The effect is large — but it may not last, and several caveats apply."

Your writing workflow

Before you write

1. **Find the claim** — one sentence.
2. Pick the **shape** — accountability, explanatory, personal.
3. Identify your strongest **number** and your best **comparison**.

While you write

1. **Headline first and last**. It disciplines the draft.
2. **Lede second**. Lead with the point; pivot by paragraph 2.
3. **Inverted pyramid** unless you have a reason not to.

After you write

1. **Cut 20%**. Then cut 10% more.
2. **Read aloud**. Mark every stumble.
3. **Hand it to someone** who has not seen your data. Can they state your claim after reading the first paragraph?

Writing about data is not a different skill from writing in general. It is writing in general — with the extra discipline of being exactly right about what the numbers show.

Ten bits of 1-minute advice

1. Small numbers



To make a very small proportion more understandable, divide it into 1.

For instance, 0.0091 (or 0.9%) is the proportion of the U.S. population who die every year (about 3.1 out of 335m).

Start from the raw ratio:

$$p = \frac{\text{deaths per year}}{\text{U.S. population}} = \frac{D}{N}$$

Inverting gives "1 in every":

$$\frac{1}{p} = \frac{N}{D}$$

Here, $\frac{1}{p} = \frac{1}{0.0091} \approx 110$. So:

1 in 110 Americans dies each year.

Your turn

Turn the following into a "1 in every _" statement.

1. About **0.014** of German adults hold a doctorate.
2. In 2023, about **2,800 people** died in road traffic in Germany. The population is **~84 million**.

2. Small numbers II



For non-tiny proportions, prefer "X in 10" or "X in 100".

"1 in 110" is technical. "X in 10" and "X in 100" are cognitive landmarks:

- **X in 10** — a group readers can picture (a classroom, a family, a team).
- **X in 100** — the natural-frequency form of a percent. "8 in 100" and "8%" are mathematically identical, but readers reason more accurately with the first, especially about risk.

Rule of thumb:

Match the denominator to the scale. For rare events, keep "1 in N". For common ones, anchor to 10 or 100.

See [Gigerenzer et al. \(2007\)](#), *Psychological Science in the Public Interest*.

Your turn

Rewrite each number in the most intuitive form — and say why you chose that denominator:

1. **37%** of German households live in rented flats.
2. **12.7%** of working-age adults receive some form of state benefit.
3. The annual risk of dying on the road in Germany is about **1 in 30,000**.

3. Rounding



*Simon
says:*

Unless your numbers are very small,
decimal points rarely matter.

"I'm a big fan of rounding." — Sarah Cohen, *Numbers in the Newsroom*

"About 37%" is almost always better than "37.24%". Trailing digits imply a precision your data usually can't support — a point *Andrew Gelman* makes repeatedly on his blog.

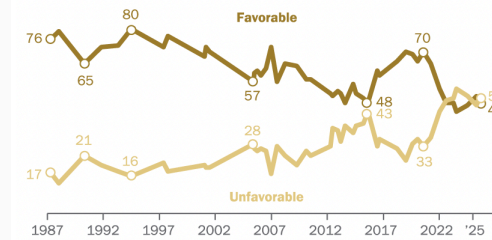
Rule of thumb:

If the next decimal doesn't change your point, drop it.

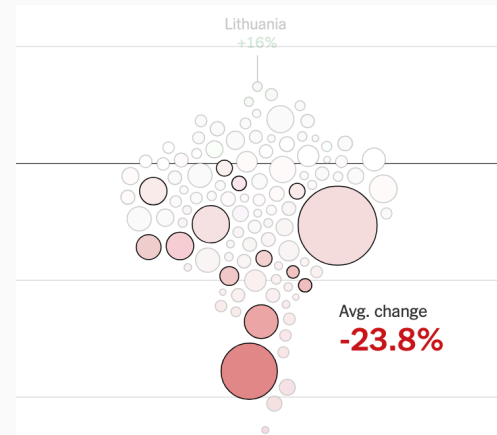
Good (**Pew Research Center**)

Favorable views of the Supreme Court remain near historic low

% who say they have a(n) ____ opinion of the Supreme Court



Bad (**NYT**, discussed by **Kaiser Fung**)



4. The 12-digit rule



**No more than 12 digits per paragraph
— and dates count.**

"2012" is 4 digits. "Between 2011 and 2012, Germany spent €47.3 billion on climate subsidies..." is already $4 + 4 + 3 = 11$ digits. You're nearly out.

Readers stop absorbing numbers long before you stop writing them. Put too many numbers into text, and you're writing a table disguised as a paragraph.

See also [Reynolds Center — "Taming Digits"](#).

Your turn — count the digits

"Between 2015 and 2023, Germany's primary energy consumption fell from 13,293 petajoules to 10,736 petajoules, while the share of renewables rose from 14.9% to 22.1% and CO₂ emissions dropped by 187 million tonnes."

1. How many digits?
2. What would you cut to get under 12?
3. What moves to a chart?

5. Avoid "X times less"



"X times less" is mathematically ambiguous.

Does it mean **one-third** the original? A 300% reduction (impossible)? Most readers don't know — and neither does half the newsroom.

Use instead:

- Fractions: *"one-third as much", "half as many"*
- Percentages: *"a 67% decrease"*

Applies to *"times smaller"*, *"times fewer"*, *"times lower"* — all the same trap.

Your turn — rewrite

Rewrite each sentence without *"times less"* or its cousins:

1. "Women in the sample were paid **1.6 times less** than men."
2. "Public-sector salaries grew **three times slower** than inflation."
3. "Rural broadband rollout has been **four times fewer** projects per capita than the urban one."

6. Percent ≠ percentage points



A rate change has two numbers. Say which you mean.

A tax rate rising from 4% to 5% is:

- **+1 percentage point** (absolute)
- **+25%** (relative)

Both are correct. They describe very different stories.

Rule: every time you report a change in a rate, say which one you mean — explicitly.

Politicians and press releases exploit this ambiguity. Don't. See [Denise-Marie Ordway's explainer](#) and this [interactive explainer](#).

Your turn

Come up with two headlines with different narratives based on the following data:

- Meanville is an average town. Its latest overall budget is \$50 million, up from \$25 million the year before.
- Last year's education budget was \$2 million. After the windfall, it shot up to \$3 million.
- Education spending as proportion of the budget was 8% last year.

7. Every number needs a denominator



Totals are meaningless without context.

Example questions to ask before publishing any number:

- **Out of how many people?** (per capita)
- **Over how long?** (per year, per decade)
- **Compared to where?** (vs. national average, vs. last year)

The denominator is the comparison that turns a number into a story.

If you can't put a denominator on it, ask whether it's a number worth reporting at all.

Your turn — add the denominators

For each bare number, write the denominator sentence you'd want before publishing:

1. "Berlin police made **5,183 drug-related arrests** last year."
2. "Fraunhofer institutes filed **512 patents** in 2024."
3. "**37,000 refugees** arrived in Lower Saxony since 2022."

8. Link your sources



*Simon
says:*

**In digital media, your footnotes are
hyperlinks. Use them.**

A print reporter writes "according to a 2023 OECD report". On the web, you can do better:

- Link **directly to the dataset** — not the press release.
- Link **the paper**, not the institution page.
- Link **the filing, the regulation, the minutes** — the primary document.

Readers who don't care won't click; readers who do will trust you more. See, e.g., [The Pudding](#) for how dense linking reads on the page.



Source [r/PeterExplainsTheJoke](#).

9. Put the methodology at the end



Lead with the finding; park the method at the bottom.

The top of your piece is for claims and stakes. The bottom is where you show your work:

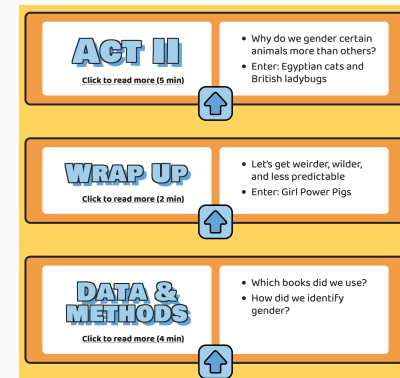
- Data sources (with links — see #8)
- What you included, excluded, and why
- Reproducible code, where possible ([GitHub](#), [Zenodo](#), [OSF](#))

Being transparent is not the same as being technical up front. Readers want the result first; sceptics will scroll down.

Now, as most of those tips, this might not apply to every piece. If your method is novel and part of the story, it may deserve a place in the lede. But if it's standard, or if your story is about the finding, not the method, put it at the end.



Source [Tagesspiegel](#).



Source [The Pudding](#).

10. Write with purpose — paragraph by paragraph



**Every paragraph should do one job.
Know what it is.**

Before you keep a paragraph, ask:

- What is this paragraph arguing?
- Does the first sentence signal the point?
- Would the piece suffer if I cut it?

A trick from research writing: read your draft **back to front**, paragraph by paragraph. If the logic still holds in reverse, your structure is sound. If it collapses, the piece is leaning on sequence, not argument.

Another trick: Just read the **first sentence of every paragraph**. Can you still understand the story? (Caveat: Sometimes you want to surprise the reader, or build suspense, or save the punchline for the end. Use your judgement.)

DIE ZEIT

NSDAP-Mitgliederkartei

Recherchieren Sie hier die NSDAP-Vergangenheit Ihrer Familie

Die ZEIT hat die Mitgliederkartei der NSDAP aufbereitet. Mit unserem Tool lassen sich Millionen Dokumente einsehen. Suchen Sie nach Ihrem Familiennamen.

Von Gregor Aisch, Jakob Bauer, Andreas Loos, Selina Rudolph, Nico Scordialo, Shairal Sharma, Christian Staas, Julian Stahnke, Sven Stockrahm und Julius Tröger

Aktualisiert am 7. April 2026, 13:02 Uhr

1.258 Verschenken Zusammenfassen

Waren Ihr Vater, Ihre Großmutter oder Ihr Ur-Opa in der NSDAP? Bis vor Kurzem ließ sich das nur über eine Anfrage beim Bundesarchiv klären oder aktuell auf der Seite des US-Nationalarchivs, das seine Mikrofilmkopien der NSDAP-Mitgliederkartei online gestellt hat. Nur sind einzelne Personen dort kaum zu finden, und die Seite selbst war unter dem Ansturm, den die Nachricht von der Veröffentlichung ausgelöst hatte, wiederholt nicht erreichbar. Die ZEIT hat deshalb den kompletten Datensatz gesichert, erstmals umfassend aufbereitet und statistisch ausgewertet. Zunächst konnten wir rund 4,5 Millionen Karteikarten aus der Nazi-Zeit direkt durchsuchbar machen. Nach einer Aktualisierung sind weitere 8,2 Millionen Dokumente recherchierbar nach Namen und Geburtsort einzelner Personen. Die Mitgliederkartei teilt sich in zwei Bestände auf:

Source **DIE ZEIT.**